

İntegrasyon Teknikleri

① Yerine Koyma (Değişken Değiştirme) Kuralı

Eğer $u=g(x)$, değer kümesi I aralığı olan türevlenebilir bir fonk. ve f de I üzerinde sürekli ise:

$$\textcircled{1} \int f(g(x)) \cdot g'(x) dx = \int f(u) du \text{ dur.}$$

$$(u=g(x) \Rightarrow du=g'(x) dx)$$

$$\textcircled{2} \int_a^b f(g(x)) \cdot g'(x) dx = \int_{g(a)}^{g(b)} f(u) du$$

$$(u=g(x) \Rightarrow du=g'(x) dx \quad \begin{array}{cc} \text{Eski sınır} & \text{Yeni Sınır} \\ x=a & \rightarrow u=g(a) \\ x=b & \rightarrow u=g(b) \end{array})$$

$$\textcircled{*} I = \int (x^3+x)^5 \cdot (3x^2+1) dx = ? \quad x^3+x=u \Rightarrow (3x^2+1) dx = du$$

$$I = \int \underbrace{(x^3+x)^5}_{u^5} \cdot \underbrace{(3x^2+1) dx}_{du} = \int u^5 du = \frac{u^6}{6} + C = \frac{(x^3+x)^6}{6} + C$$

$$\textcircled{*} \int x^2 \sin x^3 dx = ? \quad x^3=u \Rightarrow 3x^2 dx = du \Rightarrow x^2 dx = \frac{du}{3}$$

$$\int \frac{\sin x^3}{\sin u} \cdot \frac{x^2 dx}{\frac{du}{3}} = \frac{1}{3} \int \sin u du = -\frac{\cos u}{3} + C = -\frac{\cos x^3}{3} + C$$

$$\textcircled{*} \int_{-1}^1 3x^2 \sqrt{x^3+1} dx = \int_0^2 \frac{\sqrt{u}}{u^{1/2}} du = \frac{u^{3/2}}{\frac{3}{2}} \Big|_0^2 = \frac{2}{3} \cdot 2\sqrt{2} = \frac{4\sqrt{2}}{3}$$

$$\left. \begin{array}{l} x^3+1=u \\ 3x^2 dx = du \\ x=1 \Rightarrow u=2 \\ x=-1 \Rightarrow u=0 \end{array} \right\} \rightarrow$$

①

$$(*) \int x\sqrt{2x+1} dx = \int \frac{u-1}{2} \cdot \sqrt{u} \frac{du}{2} = \frac{1}{4} \int (u^{3/2} - u^{1/2}) du$$

$$\left(\begin{array}{l} 2x+1=u \\ 2dx=du \\ x=\frac{u-1}{2} \end{array} \right) \rightarrow \frac{1}{4} \left(\frac{u^{5/2}}{5/2} - \frac{u^{3/2}}{3/2} \right) + C$$

$$= \frac{1}{10} \cdot (2x+1)^{5/2} - \frac{1}{6} (2x+1)^{3/2} + C$$

$$(*) \int_0^8 \frac{\cos\sqrt{x+1}}{\sqrt{x+1}} dx = \int_1^3 2\cos u du = 2 \sin u \Big|_1^3 = 2\sin 3 - 2\sin 1$$

$$\sqrt{x+1} = u \quad \frac{dx}{2\sqrt{x+1}} = du$$

$$x=8 \Rightarrow u=3$$

$$x=0 \Rightarrow u=1$$

$$(*) \int \frac{2x}{\sqrt[3]{x^2+1}} dx = \int \frac{du}{\sqrt[3]{u}} = \int u^{-1/3} du = \frac{u^{2/3}}{2/3} + C = \frac{3}{2} (x^2+1)^{2/3} + C$$

$$x^2+1=u \rightarrow 2x dx = du$$

Tanx, Cotx, Secx, Cosecx İntegralleri

$$(*) \int \tan x dx = \int \frac{\sin x}{\cos x} dx = - \int \frac{du}{u} = -\ln|u| + C = -\ln|\cos x| + C$$

$$\cos x = u$$

↓

$$-\sin x dx = du$$

$$(*) \int \cot x dx = \int \frac{\cos x}{\sin x} dx = \int \frac{du}{u} = \ln|u| + C = \ln|\sin x| + C$$

$$\sin x = u \Rightarrow \cos x dx = du$$

$$(*) \int \sec x dx = \int \sec x \cdot \frac{(\sec x + \tan x)}{(\sec x + \tan x)} = \int \frac{\sec^2 x + \sec x \cdot \tan x}{\sec x + \tan x} dx$$

$$\sec x + \tan x = u$$

↓

$$(\sec x \tan x + \sec^2 x) dx = du$$

$$\left\{ \begin{array}{l} = \int \frac{du}{u} = \ln|u| + C \\ = \ln|\sec x + \tan x| + C \end{array} \right.$$

$$= \ln|\sec x + \tan x| + C$$

(2)

$$\textcircled{*} \int \operatorname{Cosec} x \, dx = \int \operatorname{Cosec} x \cdot \frac{(\operatorname{Cosec} x + \cot x)}{\operatorname{Cosec} x + \cot x} \, dx = \int \frac{\operatorname{Cosec}^2 x + \cot x \operatorname{Cosec} x}{\operatorname{Cosec} x + \cot x} \, dx$$

$$\begin{aligned} \operatorname{Cosec} x + \cot x &= u \\ \downarrow \\ -(\operatorname{Cosec} x \cot x + \operatorname{Cosec}^2 x) \, dx &= du \end{aligned} \quad \left\{ \begin{aligned} &= -\int \frac{du}{u} = -\ln|u| + C \\ &= -\ln|\cot x + \operatorname{Cosec} x| + C \end{aligned} \right.$$

$\sin^2 x$ ve $\cos^2 x$ İntegralleri

$\sin^2 x = \frac{1 - \cos 2x}{2}$ ve $\cos^2 x = \frac{1 + \cos 2x}{2}$ özdeşlikleri kullanılarak çözülür.

$$\textcircled{*} \int \sin^2 x \, dx = \int \frac{1 - \cos 2x}{2} \, dx = \frac{x}{2} - \frac{\sin 2x}{4} + C$$

$$\textcircled{*} \int \cos^2 x \, dx = \int \frac{1 + \cos 2x}{2} \, dx = \frac{x}{2} + \frac{\sin 2x}{4} + C$$

Simetrik fonksiyonların Belirli İntegrali

$f(x)$, $[-a, a]$ da sürekli olsun.

a) f , çift fonksiyon ise : $\int_{-a}^a f(x) \, dx = 2 \int_0^a f(x) \, dx$ dir.

(Çünkü $f(x)$ y-eksenine göre simetriktir.

$$\left. \begin{aligned} &\text{A} \quad \begin{array}{c} \text{A} \\ \text{---} \end{array} \quad \begin{array}{c} \text{A} \\ \text{---} \end{array} \\ &\text{---} \end{aligned} \right\} \Rightarrow \int_{-a}^a f(x) \, dx = 2A = 2 \int_0^a f(x) \, dx$$

b) f , tek fonk. ise : $\int_{-a}^a f(x) \, dx = 0$ dir.

(Çünkü $f(x)$ orijine göre simetriktir.

$$\left. \begin{aligned} &\text{A} \quad \begin{array}{c} \text{A} \\ \text{---} \end{array} \\ &\text{---} \end{aligned} \right\} \Rightarrow \int_{-a}^a f(x) \, dx = A - A = 0$$

Ters Trigonometrik fonksiyonları veren integraller

$$(*) \int \frac{dx}{\sqrt{1-x^2}} = \text{ArcSin} x + C$$

$$(*) \int \frac{dx}{1+x^2} = \text{ArcTan} x + C$$

$$(*) \int \frac{dx}{4+x^2} = \frac{1}{4} \int \frac{dx}{1+(\frac{x}{2})^2} = \frac{1}{2} \int \frac{du}{1+u^2} = \frac{1}{2} \text{ArcTan} u + C = \frac{1}{2} \text{ArcTan} \frac{x}{2} + C$$

$$(u = \frac{x}{2} \Rightarrow du = \frac{dx}{2})$$

KURAL:

$$(*) \int \frac{dx}{a^2+x^2} = \frac{1}{a} \text{ArcTan} \frac{x}{a} + C$$

$$(*) \int \frac{dx}{\sqrt{4-x^2}} = \int \frac{dx}{2\sqrt{1-(\frac{x}{2})^2}} = \int \frac{du}{\sqrt{1-u^2}} = \text{ArcSin} u + C = \text{ArcSin} \frac{x}{2} + C$$

$$\frac{x}{2} = u \rightarrow \frac{dx}{2} = du$$

KURAL :
$$(*) \int \frac{dx}{\sqrt{a^2-x^2}} = \text{ArcSin} \frac{x}{a} + C$$

$$(*) \int \frac{dx}{x^2+3} = \frac{1}{\sqrt{3}} \text{ArcTan} \frac{x}{\sqrt{3}} + C$$

$$(*) \int \frac{dx}{\sqrt{5-x^2}} = \text{ArcSin} \frac{x}{\sqrt{5}} + C$$

$$(*) \int \frac{dx}{x^2+8} = \frac{1}{2\sqrt{2}} \text{ArcTan} \frac{x}{2\sqrt{2}} + C$$

$$(*) \int \frac{dx}{\sqrt{16-x^2}} = \text{ArcSin} \frac{x}{4} + C$$

② Kısmi İntegrasyon

$u(x)$ ve $v(x)$ türevlenebilir fonksiyonlar olsun.

$$\frac{d}{dx}(u \cdot v) = v \cdot \frac{du}{dx} + u \cdot \frac{dv}{dx}$$

↓

$$\int \frac{d}{dx}(u \cdot v) dx = \int v \frac{du}{dx} dx + \int u \frac{dv}{dx} dx$$

↓

$$u \cdot v = \int v du + \int u dv \Rightarrow \boxed{\int u dv = u \cdot v - \int v du} \rightarrow \text{Kısmi İntegrasyon Formülü}$$

★ Belirli integralde :

$$\boxed{\int_a^b u dv = u \cdot v \Big|_a^b - \int_a^b v du}$$

★ u deme sıralaması : L A P T Ü

- Üstel fonk.
- Trigonometrik fonk.
- Polinom
- Ters Trig. fonk.
- Logaritmik fonk.

★

u	$\xrightarrow{\text{Türev}}$	du
dv	$\xrightarrow{\text{integral}}$	v

⊗ $\int x e^x dx = x e^x - \int e^x dx = x e^x - e^x + C$

$$x = u \rightarrow dx = du$$

$$e^x dx = dv \rightarrow e^x = v$$

⊗ $\int \ln x dx = x \ln x - \int dx = x \ln x - x + C$

$$\ln x = u, \quad dx = dv$$

$$\downarrow \quad \downarrow$$

$$\frac{dx}{x} = du \quad x = v$$

⊗ $\int x^2 \sin x dx = -x^2 \cos x + 2 \int x \cos x dx = -x^2 \cos x + 2x \sin x - 2 \int \sin x dx$

$$= -x^2 \cos x + 2x \sin x + 2 \cos x + C$$

$$\left. \begin{array}{l} x^2 = u \\ \downarrow \\ 2x dx = du \end{array} \right\} \begin{array}{l} \sin x dx = dv \\ \downarrow \\ -\cos x = v \end{array} \quad \left. \begin{array}{l} x = u \\ \downarrow \\ dx = du \end{array} \right\} \begin{array}{l} \cos x dx = dv \\ \downarrow \\ \sin x = v \end{array}$$

$$\textcircled{*} \int x \operatorname{Arctan} x \, dx = \frac{x^2}{2} \operatorname{Arctan} x - \frac{1}{2} \int \frac{x^2}{1+x^2} \, dx$$

$$\left. \begin{array}{l} \operatorname{Arctan} x = u \\ \downarrow \\ \frac{dx}{1+x^2} = du \\ x dx = dv \\ \downarrow \\ \frac{x^2}{2} = v \end{array} \right\} \begin{array}{l} = \frac{x^2}{2} \operatorname{Arctan} x - \frac{1}{2} \int \left(1 - \frac{1}{1+x^2}\right) dx \\ = \frac{x^2}{2} \operatorname{Arctan} x - \frac{x}{2} + \frac{\operatorname{Arctan} x}{2} + c \end{array}$$

$$\textcircled{*} \int \operatorname{ArcSin} x \, dx = x \operatorname{ArcSin} x - \int \frac{x}{\sqrt{1-x^2}} \, dx = x \operatorname{ArcSin} x + \frac{1}{2} \int \frac{dt}{\sqrt{t}}$$

$$\left. \begin{array}{l} \operatorname{ArcSin} x = u \\ \downarrow \\ \frac{dx}{\sqrt{1-x^2}} = du \\ dx = dv \\ \downarrow \\ x = v \end{array} \right\} \begin{array}{l} 1-x^2 = t \\ -2x dx = dt \end{array} \quad \begin{array}{l} = x \operatorname{ArcSin} x + \frac{1}{2} \cdot 2\sqrt{1-x^2} + c \\ = x \operatorname{ArcSin} x + \sqrt{1-x^2} + c \end{array}$$

$$\textcircled{*} I = \int e^x \cos x \, dx = e^x \cos x + \int e^x \sin x \, dx$$

$$\left. \begin{array}{l} \cos x = u \\ \downarrow \\ -\sin x dx = du \\ e^x dx = dv \\ \downarrow \\ e^x = v \end{array} \right\} \left. \begin{array}{l} \sin x = u \\ \downarrow \\ \cos x dx = du \\ e^x dx = dv \\ \downarrow \\ e^x = v \end{array} \right\}$$

$$I = e^x \cos x + e^x \sin x - \int e^x \cos x \, dx$$

I

$$2I = e^x (\cos x + \sin x) \Rightarrow I = \frac{e^x}{2} (\cos x + \sin x) + C$$

$$\textcircled{*} \int_1^e x^3 \ln^2 x \, dx = \frac{x^4}{4} \ln^2 x \Big|_1^e - \frac{1}{2} \int_1^e x^3 \ln x \, dx = \frac{e^4}{4} - \frac{x^4}{4 \cdot 2} \ln x \Big|_1^e + \frac{1}{8} \int_1^e x^3 \, dx$$

$$\left. \begin{array}{l} \ln^2 x = u \\ \frac{2 \ln x}{x} dx = du \\ x^3 dx = dv \\ \downarrow \\ \frac{x^4}{4} = v \end{array} \right\} \left. \begin{array}{l} \ln x = u \\ \downarrow \\ \frac{dx}{x} = du \\ x^3 dx = dv \\ \downarrow \\ \frac{x^4}{4} = v \end{array} \right\} \quad = \frac{e^4}{4} - \frac{e^4}{8} + \frac{x^4}{32} \Big|_1^e = \frac{5e^4 - 1}{32}$$

③ $\int \sin^m x \cdot \cos^n x dx$ integralleri

* m çift, n tek ise: $u = \sin x$
 $du = \cos x dx$

* n çift, m tek ise: $u = \cos x$
 $du = -\sin x dx$

dönüşümleri ve

$\sin^2 x + \cos^2 x = 1$ özdeşliği

kullanılarak integral çözülür.

* Hem m, hem de n tek ise herhangi birine u denir

* Hem m, hem de n çift ise: $\cos^2 x = \frac{1 + \cos 2x}{2}$
 $\sin^2 x = \frac{1 - \cos 2x}{2}$ } özdeşlikleri ile derece düşürülür

$$\begin{aligned} \textcircled{*} \int \sin^3 x \cdot \cos^3 x dx &= \int \frac{\sin^2 x}{1-u^2} \cdot \frac{\cos^2 x}{u^2} \cdot \frac{\sin x dx}{-du} = \int u^8 (u^2 - 1) du \\ \cos x = u \quad -\sin x dx &= du \\ &= \frac{u^{11}}{11} - \frac{u^9}{9} + C \\ &= \frac{(\cos x)^{11}}{11} - \frac{(\cos x)^9}{9} + C \end{aligned}$$

$$\begin{aligned} \textcircled{*} \int \cos^5(ax) dx &= \int \frac{(\cos^2(ax))^2}{(1-u^2)^2} \cdot \frac{\cos ax dx}{\frac{du}{a}} = \frac{1}{a} \int (1 - 2u^2 + u^4) du \\ \sin ax = u \quad \downarrow \\ a \cos ax dx &= du \\ &= \frac{1}{a} \left(\sin x - \frac{2}{3} \sin^3 ax - \frac{1}{5} \sin^5 ax \right) + C \end{aligned}$$

$$\begin{aligned} \textcircled{*} \int \sin^5 x \cdot \cos^3 x dx &= \int \frac{\sin^4 x}{u^5} \cdot \frac{\cos^2 x}{1-u^2} \cdot \frac{\cos x dx}{du} = \int (u^5 - u^7) du = \frac{(\sin x)^6}{6} - \frac{(\sin x)^8}{8} + C \\ \sin x = u \quad \cos x dx &= du \end{aligned}$$

$$\begin{aligned} \textcircled{*} \int \sin^4 x dx &= \int (\sin^2 x)^2 dx = \int \left(\frac{1 - \cos 2x}{2} \right)^2 dx = \frac{1}{4} \int (1 - 2\cos 2x + \frac{\cos^2 2x}{1 + \frac{\cos 4x}{2}}) dx \\ &= \frac{x}{4} - \frac{\sin 2x}{4} + \frac{1}{4} \int \frac{1 + \cos 4x}{2} dx \\ &= \frac{x}{4} - \frac{\sin 2x}{4} + \frac{x}{8} + \frac{\sin 4x}{32} + C \end{aligned}$$

④ $\int \sec^m x \cdot \tan^n x dx$ integralleri

a) $u = \sec x \rightarrow du = \sec x \tan x dx$

b) $u = \tan x \rightarrow du = \sec^2 x dx$

c) Kısmi integrasyon

Mollarından biri ve

$\sec^2 x = 1 + \tan^2 x$ özdeşliği

kullanılarak çözülür.

① $\int \tan^2 x dx = \int (\sec^2 x - 1) dx = \tan x - x + C$

② $\int \sec^4 x dx = \int \frac{\sec^2 x}{1+u^2} \cdot \frac{\sec^2 x dx}{du} = \int (1+u^2) du = u + \frac{u^3}{3} + C$

$\tan x = u$

↓

$\sec^2 x dx = du$

$= \tan x + \frac{(\tan x)^3}{3} + C$

③ $\int \sec^3 x \cdot \tan^3 x dx = \int \frac{\sec^2 x}{u^2} \cdot \frac{\tan^2 x}{u^2-1} \cdot \frac{\sec x \cdot \tan x dx}{du} = \int (u^4 - u^2) du$

$\sec x = u \rightarrow \sec x \cdot \tan x dx = du$

$= \frac{(\sec x)^5}{5} - \frac{(\sec x)^3}{3} + C$

④ $I = \int \sec^3 x dx = \int \sec x \cdot \sec^2 x dx = \sec x \tan x - \int \sec x \cdot \frac{\tan^2 x dx}{\sec^2 x - 1}$

$\sec x = u$

↓

$\sec x \tan x dx = du$

$\sec^2 x dx = dv$

↓

$v = \tan x$

$= \sec x \tan x - \underbrace{\int \sec^3 x dx}_I + \int \sec x dx$

⇓

$2I = \sec x \tan x + \ln |\sec x + \tan x|$

$I = \frac{1}{2} (\sec x \tan x + \ln |\sec x + \tan x|) + C$

⑤ $\int \sin mx \cdot \cos nx \, dx$ integralleri

$\sin mx \cdot \cos nx$, $\sin mx \cdot \sin nx$, $\cos mx \cdot \cos nx$ çarpımlarını içeren integral-
ni çözmek için:

$$\sin mx \cdot \cos nx = \frac{1}{2} (\sin(m-n)x + \sin(m+n)x)$$

$$\sin mx \cdot \sin nx = \frac{1}{2} (\cos(m-n)x - \cos(m+n)x)$$

$$\cos mx \cdot \cos nx = \frac{1}{2} (\cos(m-n)x + \cos(m+n)x)$$

dönüşümleri
kullanılır.

$$\begin{aligned} (*) \int \sin 3x \cdot \cos 5x \, dx &= \frac{1}{2} \int (\sin(3x-5x) + \sin(3x+5x)) \, dx \\ &= \frac{1}{2} \int (-\sin 2x + \sin 8x) \, dx = \frac{\cos 2x}{4} - \frac{\cos 8x}{16} + C \end{aligned}$$

⑥ İndirgeme Formülleri

(*) $\int \tan^n x \, dx$ integrali için bir indirgeme formülü bulup,
bu formül yardımıyla $\int \tan^5 x \, dx$ integralini hesaplayınız.

$$I_n = \int \tan^n x \, dx \text{ olsun.}$$

$$\begin{aligned} I_n &= \int (\tan x)^{n-2} \cdot \tan^2 x \, dx = \int (\tan x)^{n-2} (\sec^2 x - 1) \, dx \\ &= \int (\tan x)^{n-2} \cdot \sec^2 x \, dx - \int (\tan x)^{n-2} \, dx \\ &\quad (\tan x = u \Rightarrow \sec^2 x \, dx = du) \quad \underbrace{\int (\tan x)^{n-2} \, dx}_{I_{n-2}} \\ &= \int u^{n-2} \, du - I_{n-2} = \frac{u^{n-1}}{n-1} - I_{n-2} \end{aligned}$$

$$I_n = \frac{(\tan x)^{n-1}}{n-1} - I_{n-2}$$

③

$$I_n = \frac{(\tan x)^{n-1}}{n-1} - I_{n-2}$$

↓

$$\begin{aligned} I_5 &= \int (\tan x)^5 dx = \frac{(\tan x)^4}{4} - I_3 = \frac{(\tan x)^4}{4} - \left(\frac{(\tan x)^2}{2} - I_1 \right) \\ &= \frac{(\tan x)^4}{4} - \frac{(\tan x)^2}{2} + \int \tan x \\ &= \frac{(\tan x)^4}{4} - \frac{(\tan x)^2}{2} - \ln |\cos x| + C \end{aligned}$$

* $\int x^n \cdot e^{ax} dx$ ($a \neq 0$) integrali için bir indirgeme formülü bulup $\int x^3 \cdot e^{ax} dx$ integralini bu formül yardımı ile hesaplayın.

$$I_n = \int x^n e^{ax} dx \text{ olsun.} \quad \begin{aligned} x^n &= u \rightarrow n \cdot x^{n-1} \cdot dx = du \quad x = du \\ e^{ax} dx &= dv \rightarrow \frac{e^{ax}}{a} = v \end{aligned}$$

$$I_n = \frac{1}{a} \cdot x^n \cdot e^{ax} - \frac{n}{a} \int x^{n-1} \cdot e^{ax} dx = \frac{1}{a} \cdot x^n \cdot e^{ax} - \frac{n}{a} I_{n-1}$$

$$\boxed{I_n = \frac{1}{a} \cdot x^n e^{ax} - \frac{n}{a} I_{n-1}}$$

$$I_3 = \int x^3 e^{ax} dx = \frac{1}{a} x^3 e^{ax} - \frac{3}{a} I_2 = \frac{1}{a} x^3 e^{ax} - \frac{3}{a} \left(\frac{1}{a} x^2 e^{ax} - \frac{2}{a} I_1 \right)$$

$$= \frac{1}{a} x^3 e^{ax} - \frac{3}{a^2} x^2 e^{ax} + \frac{6}{a^2} \left(\frac{1}{a} x e^{ax} - \frac{1}{a} I_0 \right)$$

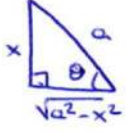
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$$= \frac{1}{a} x^3 e^{ax} - \frac{3}{a^2} x^2 e^{ax} + \frac{6}{a^3} x e^{ax} - \frac{6}{a^3} \int e^{ax} dx$$

$$= \frac{1}{a} x^3 e^{ax} - \frac{3}{a^2} x^2 e^{ax} + \frac{6}{a^3} x e^{ax} - \frac{6}{a^4} e^{ax} + C$$

⑦ Trigonometrik Değişken Dönüşümleri

$x = a \sin \theta$ dönüşümü:



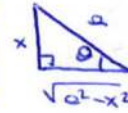
★ $\theta = \text{ArcSin} \frac{x}{a}$ dönüşümüne denktir

★ $\sqrt{a^2 - x^2}$ ($a > 0$) seklinde terim içeren integrallerde kullanılır.

★ $\sqrt{a^2 - x^2}$ ifadesi $-a \leq x \leq a$ için anlamlıdır; bu ise $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$ için mümkündür.

★ $x = a \sin \theta$ dönüşümü ile:

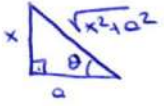
$$\sqrt{a^2 - x^2} = \sqrt{a^2 - a^2 \sin^2 \theta} = a |\cos \theta| = a \cos \theta$$
$$x = a \sin \theta \quad dx = a \cos \theta d\theta$$



$$\Rightarrow \cos \theta = \frac{\sqrt{a^2 - x^2}}{a}$$

$$\tan \theta = \frac{x}{\sqrt{a^2 - x^2}}$$

$x = a \tan \theta$ dönüşümü



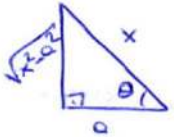
★ $\theta = \text{ArcTan} \frac{x}{a}$ dönüşümüne denktir

★ $\sqrt{a^2 + x^2}$ veya $a^2 + x^2$ ($a > 0$) içeren integrallerde kullanılır. $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$ için:

$$\sqrt{a^2 + x^2} = \sqrt{a^2 + a^2 \tan^2 \theta} = a |\sec \theta| = a \sec \theta$$

$$x = a \tan \theta \rightarrow dx = a \sec^2 \theta d\theta$$

$x = a \sec \theta$ dönüşümü



★ $\theta = \text{ArcSec} \frac{x}{a}$ dönüşümüne denktir

★ $\sqrt{x^2 - a^2}$ içeren integrallerde kullanılır

★ $\text{ArcSec} \theta$ nin tanım kümesi: $0 \leq \theta < \frac{\pi}{2}$ ve $\frac{\pi}{2} < \theta \leq \pi$ dir.

işlem kolaylığı için $0 \leq \theta < \frac{\pi}{2}$ seçersek:

$$\sqrt{x^2 - a^2} = \sqrt{a^2 \sec^2 \theta - a^2} = a |\tan \theta| = a \tan \theta$$

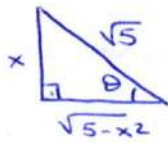
$$x = a \sec \theta \rightarrow dx = a \sec \theta \tan \theta d\theta$$

$$(*) \int \frac{dx}{(5-x^2)^{3/2}} = \int \frac{\sqrt{5} \cos \theta d\theta}{5\sqrt{5} \cos^3 \theta} = \frac{1}{5} \int \sec^2 \theta d\theta = \frac{1}{5} \tan \theta + c$$

$$x = \sqrt{5} \sin \theta$$

$$dx = \sqrt{5} \cos \theta d\theta$$

$$\sqrt{5-x^2} = \sqrt{5(1-\sin^2 \theta)} = \sqrt{5} \cos \theta$$



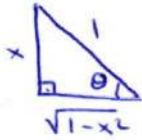
$$= \frac{1}{5} \frac{x}{\sqrt{5-x^2}} + c$$

$$(*) \int \frac{x^2}{\sqrt{1-x^2}} dx = \int \frac{\sin^2 \theta}{\cos \theta} \cancel{\cos \theta} d\theta = \int \sin^2 \theta d\theta = \int \frac{1-\cos 2\theta}{2} d\theta$$

$$x = \sin \theta$$

$$dx = \cos \theta d\theta$$

$$\sqrt{1-x^2} = \cos \theta$$



$$= \frac{\theta}{2} - \frac{\sin 2\theta}{4} + c$$

$$= \frac{\theta}{2} - \frac{\sin \theta \cdot \cos \theta}{2} + c$$

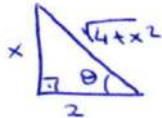
$$= \frac{\text{ArcSin} x}{2} - \frac{1}{2} \cdot x \cdot \sqrt{1-x^2} + c$$

$$(*) \int \frac{dx}{\sqrt{4+x^2}} = \int \frac{2 \sec^2 \theta d\theta}{2 \sec \theta} = \int \sec \theta d\theta = \ln |\sec \theta + \tan \theta| + c$$

$$x = 2 \tan \theta$$

$$dx = 2 \sec^2 \theta d\theta$$

$$\sqrt{4+x^2} = 2 \sec \theta$$



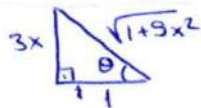
$$= \ln \left| \frac{\sqrt{4+x^2}}{2} + \frac{x}{2} \right| + c$$

$$(*) \int \frac{dx}{(1+9x^2)^2} = \int \frac{1}{3} \cdot \frac{\sec^2 \theta}{\sec^4 \theta} d\theta = \frac{1}{3} \int \cos^2 \theta d\theta = \frac{1}{3} \int \frac{1+\cos 2\theta}{2} d\theta$$

$$3x = \tan \theta$$

$$3dx = \sec^2 \theta d\theta$$

$$1+9x^2 = \sec^2 \theta$$



$$= \frac{1}{6} \left(\theta + \frac{\sin 2\theta}{2} \right) + c$$

$$= \frac{\theta}{6} + \frac{1}{6} \sin \theta \cos \theta + c$$

$$= \frac{\text{Arc Tan} x}{6} + \frac{1}{6} \cdot \frac{3x}{\sqrt{1+9x^2}} \cdot \frac{1}{\sqrt{1+9x^2}} + c$$

$$(*) \int \frac{dx}{x\sqrt{x^2-4}} = \int \frac{2 \sec \theta \tan \theta}{2 \sec \theta \cdot 2 \tan \theta} d\theta = \int \frac{d\theta}{2} = \frac{1}{2} \theta + c = \frac{1}{2} \text{ArcSec} \frac{x}{2} + c$$

$$x = 2 \sec \theta$$

$$dx = 2 \sec \theta \tan \theta d\theta$$

$$\sqrt{x^2-4} = \sqrt{4(\sec^2 \theta - 1)} = 2 \tan \theta$$

⑧ Rasyonel Fonksiyonların Basit Kesirlerle İntegrasyonu

Rasyonel fonksiyonların daha basit kesirlerin toplamı olarak yazma işlemine "Basit Kesirler Yöntemi" denir.

$$\frac{5x-3}{x^2-2x-3} = \frac{A}{x+1} + \frac{B}{x-3} \quad A, B \rightarrow \text{Belirsiz Katsayılar}$$

* $\frac{P(x)}{Q(x)}$ rasyonel fonksiyonunu basit kesirlere ayırabilmek için: $d(P(x)) < d(Q(x))$ olmalıdır. Değilse polinom bölmesi ile bu hale getirilir.

Basit Kesirlere Ayırma Yöntemi:

① $x-r$, $Q(x)$ in lineer (1. dereceden) çarpanı olsun.
 $(x-r)^m$, $Q(x)$ i bölen $(x-r)$ nin en büyük kuvveti olsun.
O zaman bu çarpan için m tane kısmi kesirin toplamı yazılır:

$$(x-r)^m \text{ çarpanı için: } \frac{A_1}{x-r} + \frac{A_2}{(x-r)^2} + \dots + \frac{A_m}{(x-r)^m}$$

Bu işlem, $Q(x)$ in her lineer çarpanı için yapılır.

② x^2+px+q , $Q(x)$ in kuadratik (2. dereceden) çarpanı olsun.
 $(x^2+px+q)^n$, bu çarpanın $Q(x)$ i bölen en büyük kuvveti olsun

$$(x^2+px+q)^n \text{ için: } \frac{B_1x+C_1}{x^2+px+q} + \frac{B_2x+C_2}{(x^2+px+q)^2} + \dots + \frac{B_nx+C_n}{(x^2+px+q)^n} \quad \text{Kısmi kesirleri yazılır.}$$

③ Orijinal $\frac{P(x)}{Q(x)}$ kesri, tüm bu basit kesirlerin toplamına

esitlenir. Paydalar eşitlenip, paylar da polinom eşitliği yardımıyla katsayılar hesaplanır.

* Kısmi (Heaviside) Yöntemi

$d(P(x)) < d(Q(x))$ ve $Q(x) = (x-r_1)(x-r_2)\dots(x-r_n)$, $Q(x)$ in birbirinden farklı, kuvveti 1 olan lineer çarpanları ise:

$$\frac{P(x)}{Q(x)} = \frac{P(x)}{(x-r_1)(x-r_2)\dots(x-r_n)} = \frac{A_1}{x-r_1} + \frac{A_2}{x-r_2} + \dots + \frac{A_n}{x-r_n} \quad \text{olur ve}$$

$$A_i = \lim_{x \rightarrow r_i} (x-r_i) \cdot \frac{P(x)}{(x-r_1)\dots(x-r_n)}$$

dir. (Pratikte bunu, $\frac{P(x)}{Q(x)}$ in paydasındaki $x-r_i$ çarpanını çıkarıp kalan ifadede $x=r_i$ yazarak yaparız.)

$$\textcircled{*} \int \frac{5x-3}{x^2-2x-3} dx = ?$$

$$\frac{5x-3}{(x-3)(x+1)} = \frac{A}{x-3} + \frac{B}{x+1}$$

$$A = \left. \frac{5x-3}{x+1} \right|_{x=3} = 3 \quad B = \left. \frac{5x-3}{x-3} \right|_{x=-1} = 2$$

$$\int \frac{5x-3}{x^2-2x-3} dx = \int \left(\frac{3}{x-3} + \frac{2}{x+1} \right) dx = 3 \ln|x-3| + 2 \ln|x+1| + C$$

$$\textcircled{*} I = \int \frac{x^2+4x+1}{(x-1)(x+1)(x+3)} dx = ?$$

$$\frac{x^2+4x+1}{(x-1)(x+1)(x+3)} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{C}{x+3}$$

$$A = \left. \frac{x^2+4x+1}{(x+1)(x+3)} \right|_{x=1} = \frac{6}{2 \cdot 4} = \frac{3}{4}$$

$$B = \left. \frac{x^2+4x+1}{(x-1)(x+3)} \right|_{x=-1} = \frac{-2}{-2 \cdot 2} = \frac{1}{2}$$

$$C = \left. \frac{x^2+4x+1}{(x-1)(x+1)} \right|_{x=-3} = \frac{9-12+1}{-4 \cdot -2} = -\frac{1}{4}$$

$$I = \int \left(\frac{3}{4} \cdot \frac{1}{x-1} + \frac{1}{2} \cdot \frac{1}{x+1} + \frac{1}{4} \cdot \frac{1}{x+3} \right) dx$$

$$= \frac{3}{4} \ln|x-1| + \frac{1}{2} \ln|x+1| - \frac{1}{4} \ln|x+3| + C$$

$$(*) \int \frac{6x+7}{(x+2)^2} dx = ?$$

$$\frac{6x+7}{(x+2)^2} = \frac{A}{x+2} + \frac{B}{(x+2)^2} = \frac{Ax+2A+B}{(x+2)^2}$$

$$\boxed{A=6}$$

$$2A+B=7 \Rightarrow \boxed{B=-5}$$

$$\int \frac{6x+7}{(x+2)^2} dx = \int \left(\frac{6}{x+2} - \frac{5}{(x+2)^2} \right) dx = 6 \ln|x+2| + \frac{5}{x+2} + C$$

$$(*) \int \frac{x^3-4x^2-4x-3}{x^2-4x-5} dx = ? \quad \frac{x(x^2-4x-5)}{x^3-4x^2-4x-x+3} = x + \frac{x-3}{x^2-4x-5}$$

$$\frac{x-3}{x^2-4x-5} = \frac{A}{x-5} + \frac{B}{x+1} \Rightarrow A = \frac{1}{3} \quad B = \frac{2}{3}$$

$$I = \int \left(x + \frac{1}{3} \cdot \frac{1}{x-3} + \frac{2}{3} \cdot \frac{1}{x+1} \right) dx = \frac{x^2}{2} + \frac{1}{3} \ln|x-3| + \frac{2}{3} \ln|x+1| + C$$

$$(*) I = \int \frac{4-2x}{(x^2+1)(x-1)} dx = ?$$

$$\frac{4-2x}{(x^2+1)(x-1)} = \frac{Ax+C}{x^2+1} + \frac{B}{x-1} = \frac{Ax^2-Ax+Bx^2+B+Cx-C}{(x-1)(x^2+1)} \Rightarrow$$

$$A+B=0$$

$$C-A=-2$$

$$B-C=4$$

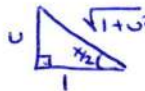
$$B=1 \quad A=-1 \quad C=-3$$

$$I = \int \left(-\frac{x+3}{x^2+1} + \frac{1}{x-1} \right) dx = \int -\frac{x}{x^2+1} dx + \int \frac{3}{x^2+1} dx + \int \frac{dx}{x-1}$$

$$= -\frac{1}{2} \ln|x^2+1| + 3 \operatorname{Arctan} x + \ln|x-1| + C$$

9) $\tan \frac{x}{2}$ Dönüşümü

$\sin x$ veya $\cos x$ içeren kesirli fonksiyonların integralini alırken kullanılır.

$\tan \frac{x}{2} = u$ dönüşümü ile: 

$$x = 2 \arctan u \Rightarrow dx = \frac{2}{1+u^2} du \text{ olur.}$$

$$\sin x = 2 \cdot \sin \frac{x}{2} \cdot \cos \frac{x}{2} = 2 \cdot \frac{u}{\sqrt{1+u^2}} \cdot \frac{1}{\sqrt{1+u^2}} = \frac{2u}{1+u^2}$$

$$\cos x = \cos^2 \frac{x}{2} - \sin^2 \frac{x}{2} = \frac{1}{1+u^2} - \frac{u^2}{1+u^2} = \frac{1-u^2}{1+u^2}$$

ÖZET

$$u = \tan \frac{x}{2} \text{ ile:}$$

$$dx = \frac{2 du}{1+u^2}$$

$$\sin x = \frac{2u}{1+u^2}$$

$$\cos x = \frac{1-u^2}{1+u^2}$$

$$*) \int \frac{dx}{1+\cos x} = \int \frac{\frac{2 du}{1+u^2}}{1+\frac{1-u^2}{1+u^2}} = \int \frac{\frac{2}{1+u^2}}{\frac{2}{1+u^2}} du = \int du = u + C = \tan \frac{x}{2} + C$$

$$\tan \frac{x}{2} = u \quad dx = \frac{2 du}{1+u^2}$$

$$\cos x = \frac{1-u^2}{1+u^2}$$

$$*) \int \frac{dx}{1-\sin x + \cos x} = \int \frac{\frac{2 du}{1+u^2}}{\frac{2(1-u)}{1+u^2}} = \int \frac{du}{1-u} = -\ln|1-u| + C = -\ln|1-\tan \frac{x}{2}| + C$$

$$\tan \frac{x}{2} = u \quad dx = \frac{2 du}{1+u^2}$$

$$1-\sin x + \cos x = 1 - \frac{2u}{1+u^2} + \frac{1-u^2}{1+u^2} = \frac{2(1-u)}{1+u^2}$$